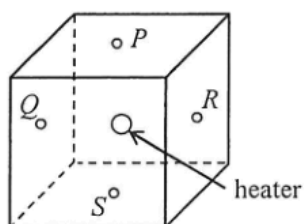


1. A heater is installed at the centre of a fully filled cubic water tank. Temperature sensors P , Q , R and S are fixed at the respective centres of the top, left, right and bottom surfaces of the tank.



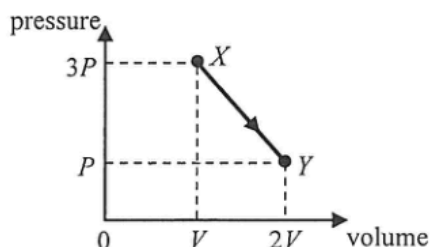
After the heater is switched on for a **short duration**, which pair of sensors below would indicate the largest temperature difference ?

- A. Q and R
- B. R and S
- C. Q and S
- D. P and R

2. An electric kettle which contains 1 kg of water at room temperature takes 168 s to heat up the water to boiling point. The kettle's rated value is '220 V, 2000 W'. Assume that all the electrical energy consumed by the kettle is transferred to the water. Which of the following statements is/are correct ?
Given: specific heat capacity of water = $4200 \text{ J kg}^{-1} \text{ }^{\circ}\text{C}^{-1}$

- (1) The initial temperature of the water is 20°C .
 - (2) The resistance of the kettle's heating element is about 24Ω .
 - (3) If the electric kettle is operated with 110 V, the time taken to heat up the water to boiling point will be doubled.
- A. (1) only
 - B. (3) only
 - C. (1) and (2) only
 - D. (1), (2) and (3)

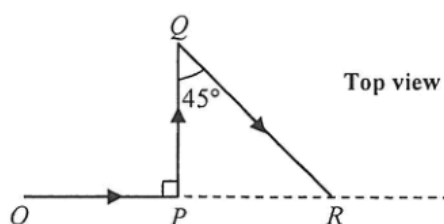
- *3. A fixed mass of an ideal gas expands from state X to state Y through a process as represented in the pressure-volume graph below.



If the temperature of the gas at state Y is 25°C , what is its temperature at state X ?

- A. -74.3°C
- B. 16.7°C
- C. 37.5°C
- D. 174°C

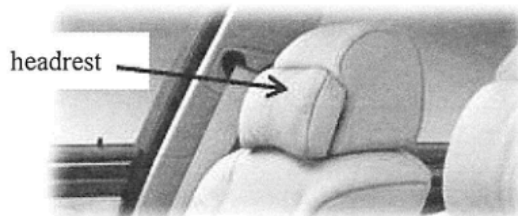
4.



A car takes 8 minutes to travel along a path $OPQR$ on a horizontal surface as shown. Given that $OP = PQ = 2$ km, find the magnitude of the average velocity of the car in this journey.

- A. 30 km h^{-1}
- B. 36 km h^{-1}
- C. 41 km h^{-1}
- D. 51 km h^{-1}

5.



For a car travelling on a highway, which of the following statements about the safety design of the headrest is/are correct ?

- (1) As the headrest is soft, it can reduce the force exerted on the passenger's head during impact.
- (2) It can minimise injury of the passenger when the car is struck by another one from behind.
- (3) It can minimise injury of the passenger when the car brakes suddenly.

- A. (1) only
- B. (3) only
- C. (1) and (2) only
- D. (2) and (3) only

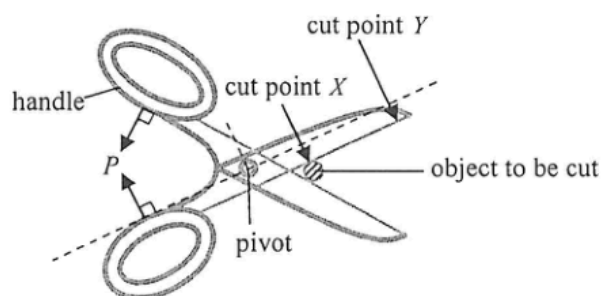
6. A particle is thrown vertically upward and its path is shown below. H is the highest point that the particle reached. Which of the following statements about the particle is/are correct ? Neglect air resistance.

- (1) Its acceleration at M is upward.
- (2) Its acceleration at H is zero.
- (3) Its acceleration at N is downward.

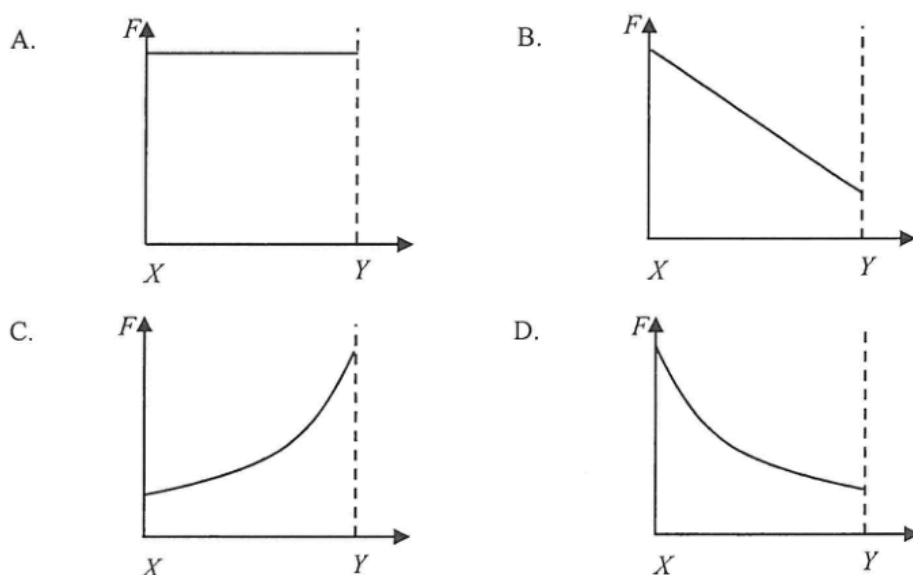
- A. (1) only
- B. (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)



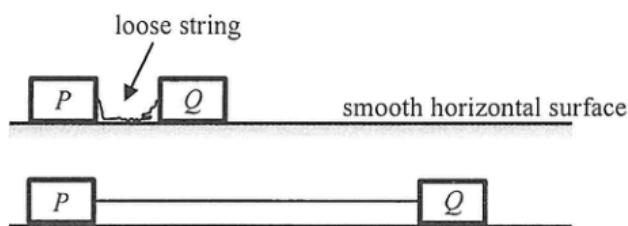
7. The figure shows that a pair of forces P of constant magnitude is applied at right angles to the handles of a pair of scissors in order to cut an object.



Which graph below best shows the variation of force F produced at cut point from X to Y when the pair of scissors is closed?



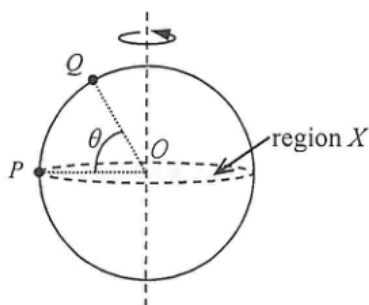
8. On a smooth horizontal surface, two identical blocks P and Q are connected by a light inextensible string. The blocks are at rest and the string is loose initially.



Q is given a speed of 4 m s^{-1} and moves to the right. Find the speeds of the blocks when the string just becomes taut and P starts to move.

	block P	block Q
A.	1 m s^{-1}	1 m s^{-1}
B.	2 m s^{-1}	1 m s^{-1}
C.	2 m s^{-1}	2 m s^{-1}
D.	4 m s^{-1}	2 m s^{-1}

- *9. Particles P and Q are fixed on the surface of a sphere rotating about a vertical axis passing through the centre O of the sphere as shown. The horizontal shaded region X divides the sphere into two halves. P is at the edge of region X while Q is at an angle of elevation θ above region X .



Find the ratio of the centripetal acceleration of P to that of Q .

- A. $1 : \cos \theta$
- B. $1 : \sin \theta$
- C. $\cos \theta : 1$
- D. $\sin \theta : 1$

2020 Q10

- *10. The diameter of Neptune is about 4 times that of the Earth and its mass is about 17 times that of the Earth. Estimate the acceleration due to gravity on Neptune's surface.
Given: acceleration due to gravity on Earth's surface $g = 9.81 \text{ m s}^{-2}$

- A. 2.3 m s^{-2}
- B. 9.2 m s^{-2}
- C. 10.4 m s^{-2}
- D. 41.7 m s^{-2}

2020 Q11

11.

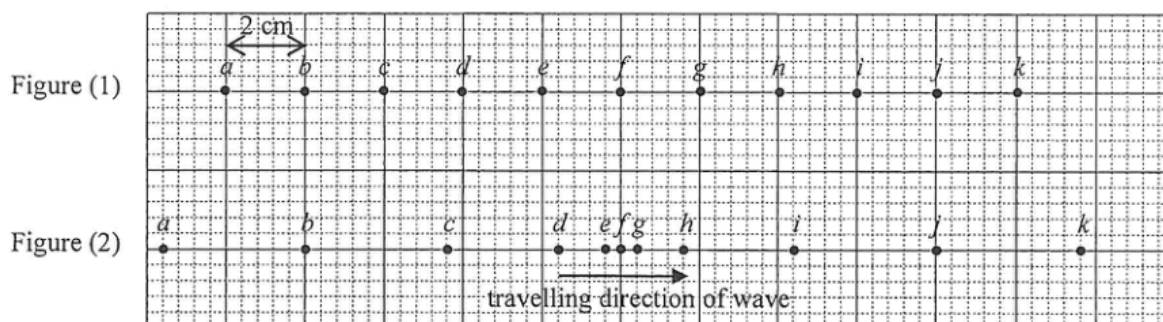
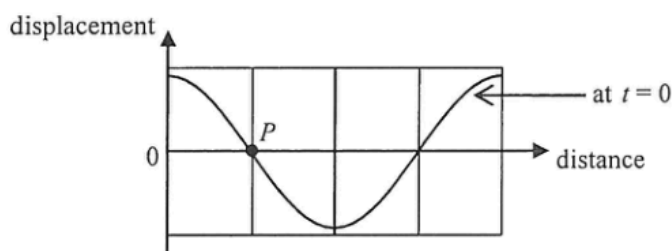


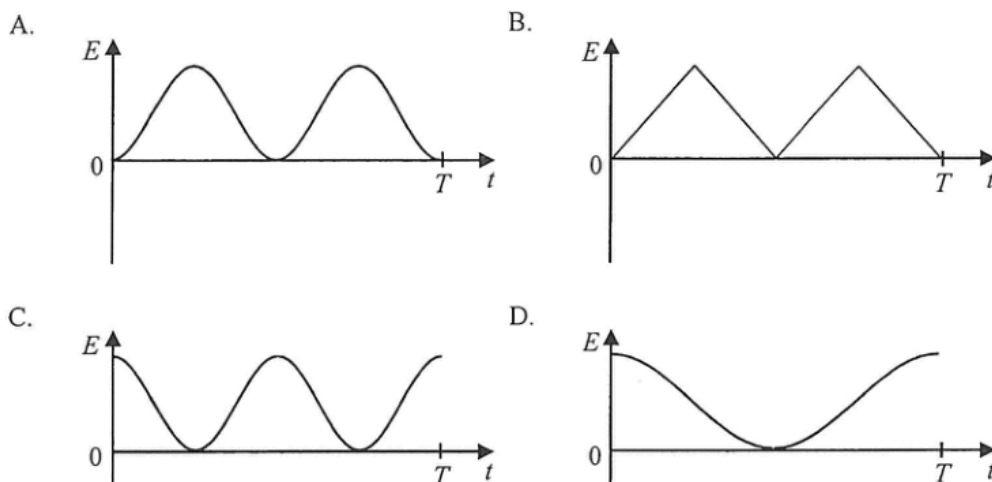
Figure (1) shows the equilibrium positions of particles a to k in a medium. The particles are separated by 2 cm from each other. A longitudinal wave of frequency 5 Hz is travelling from left to right. At a certain instant, the positions of the particles are shown in Figure (2). Determine the amplitude and speed of the wave.

- | | amplitude | speed |
|----|-----------|------------------------|
| A. | 3.6 cm | 40 cm s^{-1} |
| B. | 3.6 cm | 80 cm s^{-1} |
| C. | 2.4 cm | 40 cm s^{-1} |
| D. | 2.4 cm | 80 cm s^{-1} |

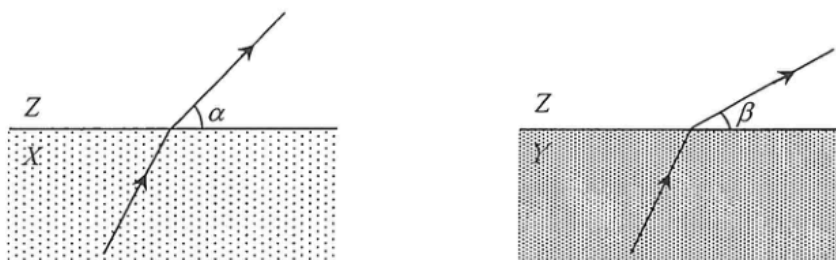
12. The figure shows part of the displacement-distance graph of a travelling wave of period T at time $t = 0$. P is a particle on the wave.



Which graph below correctly shows the variation of the particle's kinetic energy E within a period starting from $t = 0$?



13. Monochromatic light travels with the same incident angle from media X and Y respectively to another medium Z as shown.



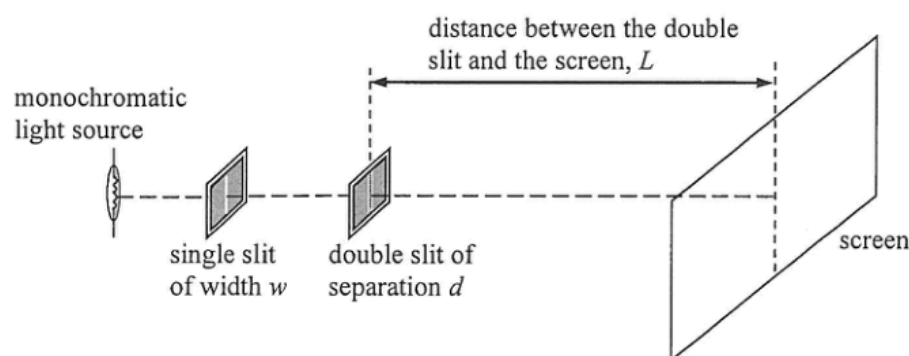
The corresponding refracted rays in Z make angles α and β respectively with the boundary plane (with $\alpha > \beta$). Which medium, X or Y , has a greater refractive index? In which medium, X or Y , does light travel faster?

	medium with a greater refractive index	medium in which light travels faster
A.	X	X
B.	X	Y
C.	Y	X
D.	Y	Y

- *14. A light beam consisting of wavelengths λ_1 and λ_2 is incident normally on a diffraction grating. The third-order diffraction of wavelength λ_1 coincides with the fourth-order diffraction of wavelength λ_2 in the resulting pattern. If λ_1 is 680 nm, find λ_2 .

- A. 510 nm
 B. 680 nm
 C. 907 nm
 D. It cannot be determined because the grating spacing is unknown.

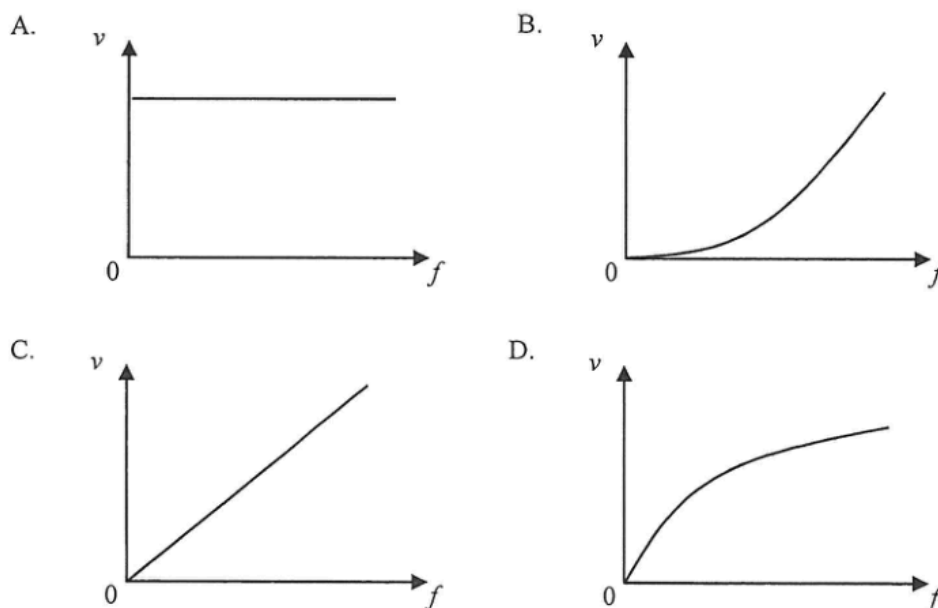
15. The figure shows a typical set-up of Young's double slit experiment.



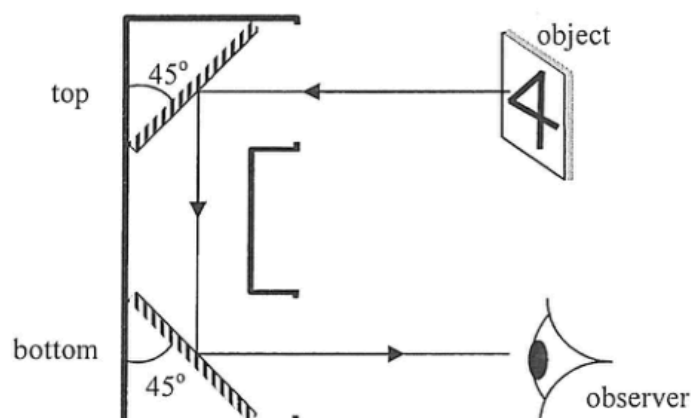
Which combination below is the best setting for displaying an observable fringe pattern on the screen ?

	w	d	L
A.	0.1 mm	1 mm	10 m
B.	0.1 mm	1 mm	1 m
C.	1 mm	0.1 mm	1 m
D.	1 mm	0.1 mm	0.1 m

16. A transverse wave propagates along a stretched string. Which graph below correctly shows the variation of the speed v of the wave with its frequency f ?



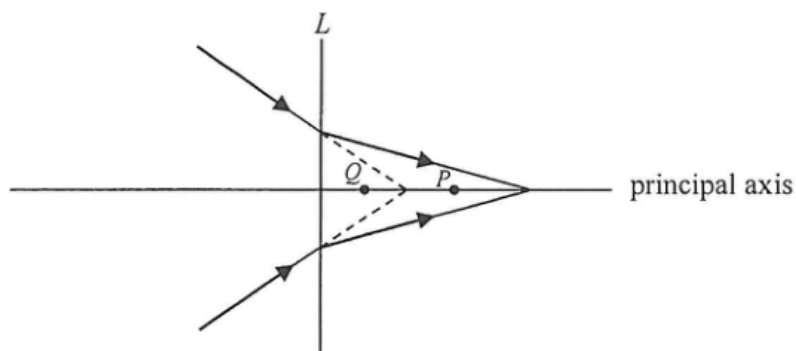
17. The figure shows a periscope designed by a student. An object is observed via the periscope.



Which image will the observer see ?

- A.
- B.
- C.
- D.

18.



Referring to the above ray diagram, what kind of lens is represented by L ? Which point, P or Q be its focus ?

	lens L	focus
A.	concave	P
B.	convex	P
C.	concave	Q
D.	convex	Q

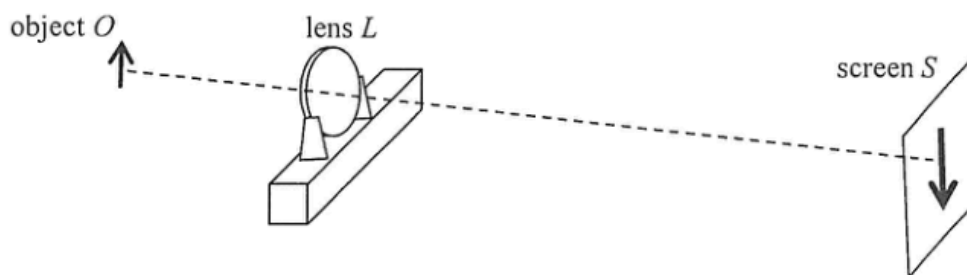
2020 Q19

19. Which of the following phenomena provides conclusive evidence that sound is a wave ?

- (1) reflection of sound from a wall
- (2) refraction of sound at the boundary between two media
- (3) interference of sound

- A. (2) only
- B. (3) only
- C. (1) and (2) only
- D. (1) and (3) only

20. The figure shows an enlarged sharp image of an object O formed on a screen S by a convex lens



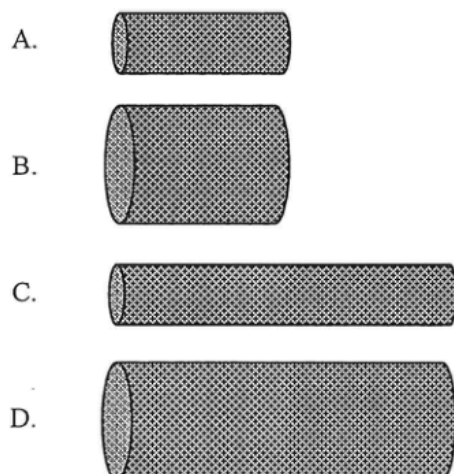
Which of the following can give a diminished sharp image on the screen ?

- (1) Keeping the positions of O and L unchanged, move S suitably closer to L .
 - (2) Keeping the positions of L and S unchanged, move O suitably farther away from L .
 - (3) Keeping the positions of O and S unchanged, move L suitably closer to S .
- A. (1) only
 B. (3) only
 C. (1) and (2) only
 D. (2) and (3) only

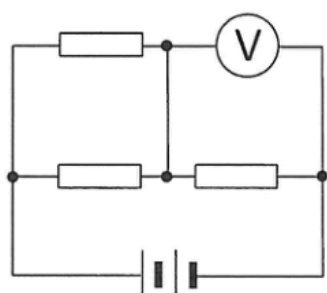
21. Which of the following statements about ultrasound is/are correct ?

- (1) Ultrasound has a shorter wavelength than audible sound.
 - (2) Ultrasound cannot be produced by vibrating objects.
 - (3) Ultrasound cannot be heard as it cannot travel through air.
- A. (1) only
 B. (3) only
 C. (1) and (2) only
 D. (2) and (3) only

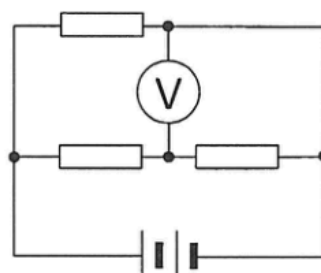
22. The cylindrical resistors below are made from the same metal. Which one would produce the most power when the same voltage is applied in turns across the two ends of each resistor ?



23. Three identical resistors, a battery of negligible internal resistance and an ideal voltmeter are connected to form Circuits (a) and (b) respectively.



Circuit (a)

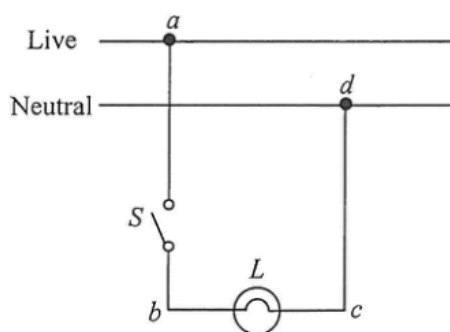


Circuit (b)

Given that the voltmeter reading is 8 V in Circuit (a), what is the voltmeter reading in Circuit (b) ?

- A. 4 V
 B. 6 V
 C. 8 V
 D. 12 V

24. The figure shows part of a domestic lighting circuit in which the bulb L does not light up when switch S is closed.



The circuit is then checked with switch S closed. Using a voltage tester to touch points b and c in turns, the tester indicates that both points are at high voltage. When touching points a and d in turns, the tester indicates only point a is at high voltage. Which of the following can be a reason for the fault?

- A. The switch S has been damaged.
- B. The filament of bulb L has been burnt out and becomes an open circuit.
- C. There is a short circuit between a and d .
- D. There is an open circuit between c and d .

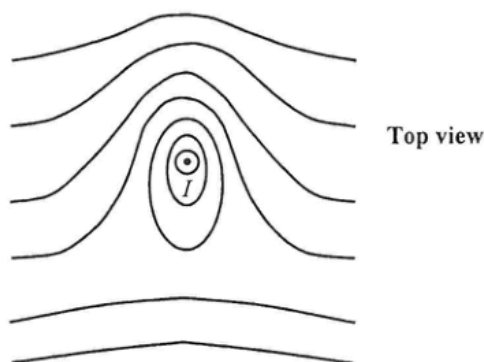
25.



The battery shown has a capacity of 1100 mA h. How much energy is delivered when the battery operates normally at a current of 250 mA for one hour? Assume that the battery's operating voltage remains at 3.7 V during that period.

- A. $(3.7 \times \frac{250}{1000} \times 3600) \text{ J}$
- B. $(3.7 \times \frac{1100}{1000} \times 3600) \text{ J}$
- C. $(3.7 \times \frac{250}{1000} \times 1) \text{ J}$
- D. $(3.7 \times \frac{1100}{1000} \times 1) \text{ J}$

26. The figure below shows the magnetic field pattern on a horizontal surface around a long vertical straight wire carrying a steady current I pointing out of the paper. The Earth's magnetic field is NOT neglected.

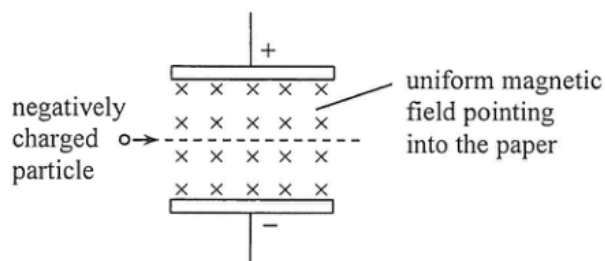


What are the directions of the following ?

	the horizontal component of the Earth's magnetic field	the magnetic force experienced by the current-carrying wire
A.	←	↓
B.	←	↑
C.	→	↓
D.	→	↑

2020 Q27

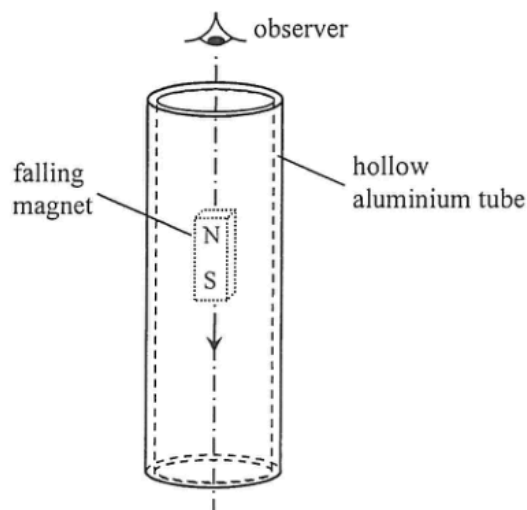
*27.



A negatively charged particle goes undeflected through a region in which a uniform electric field and a uniform magnetic field are set up as shown. The electric field is set up by the potential difference across the two parallel metal plates. Which of the following changes may cause the charged particle to deflect downward ? Neglect the effects of gravity.

- (1) increasing the potential difference across the plates
 - (2) increasing the magnitude of the charge on the particle
 - (3) increasing the particle's speed entering the region
- A. (1) only
 B. (3) only
 C. (1) and (2) only
 D. (2) and (3) only

28.

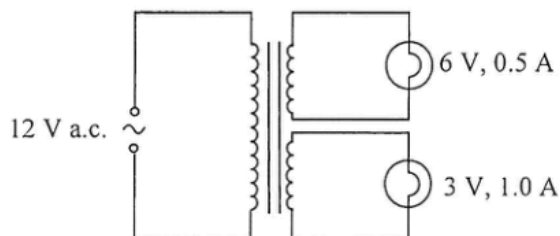


When a small strong magnet falls through a hollow aluminium tube as shown, eddy currents are induced. Which of the following correctly describes the direction of current induced in the tube when viewed by an observer from above?

- A. clockwise both above and below the magnet
- B. anti-clockwise both above and below the magnet
- C. clockwise above the magnet and anti-clockwise below the magnet
- D. anti-clockwise above the magnet and clockwise below the magnet

2020 Q29

*29.



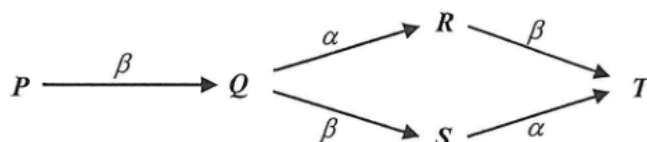
The figure shows an ideal transformer with two secondary coils connected to two light bulbs marked '6 V, 0.5 A' and '3 V, 1.0 A' respectively. When a 12 V a.c. supply is connected to the primary coil, the bulbs work at their respective rated values. Estimate the current in the primary coil.

- A. 0.25 A
- B. 0.50 A
- C. 0.75 A
- D. 1.0 A

30. The background count rate in an experiment is determined using a GM counter. Four readings of the count rate in each minute are taken. Which set of readings below is the most probable ?

	1 st minute	2 nd minute	3 rd minute	4 th minute
A.	5	62	8	69
B.	40	40	40	40
C.	60	50	30	20
D.	29	26	31	35

31.



Nuclide P can decay into nuclide T through either process $P - Q - R - T$ or process $P - Q - S - T$ as shown. Which deductions below are correct ?

- (1) P and T are isotopes of the same element.
- (2) Q and S have the same number of protons.
- (3) S has one more neutron than R .

- A. (1) and (2) only
- B. (1) and (3) only
- C. (2) and (3) only
- D. (1), (2) and (3)

- *32. The decay constant of a radioisotope of an element

- A. is random.
- B. depends on pressure and temperature.
- C. is directly proportional to the number of nucleons in the isotope.
- D. is an identifying characteristic of that isotope.

33. Two radioactive samples P and Q are freshly prepared. It is found that when $\frac{15}{16}$ of all the nuclei of P have decayed, $\frac{63}{64}$ of all the nuclei of Q have also decayed. Find the ratio $\frac{\text{half-life of } P}{\text{half-life of } Q}$.

- A. 1 : 4
- B. 2 : 3
- C. 3 : 2
- D. 4 : 1